

Efficient Unemployment and Unemployment Gap

Pascal Michailat

<https://pascalmichailat.org/v/>



CBO . natural rate of unemployment

Taking a trend of unemployment rate + adjustments

Premise on average, labor market is efficient

Problem No guarantee that labor market is efficient
on average $\rightarrow$ in matching model,

no reason to believe that labor market is efficient.

Phillips-curve approach accelerationist Phillips curve

(Friedman) . target unemployment rate such
that inflation remains constant

Problem

- care about other things than keeping inflation constant
- complete disconnect between inflation & unemployment

Efficient unemployment rate in matching model

Efficient maximizes social welfare .

Social welfare sum of welfare of all individuals

Assumption Cobb-Douglas ^{utility} matching function

Assumptions to simplify social welfare

- Linear utility function over consumption (risk neutral)
 - all individuals value consumption the same
 - can compute aggregate utility from consumption by aggregating consumption = output -
 - Disutility from work = disutility from searching for a job → value of time is the same for employed & unemployed workers
 - value of time is not relevant for welfare.
- ⇒ social welfare is determined solely by aggregate consumption = aggregate output -

Definition efficient unemployment rate is the unemployment rate that maximizes output -

Social planner - benevolent government that can allocate workers between unemployment, producing, & recruiting in order to maximize welfare = output

Social planner is subject to matching function, production function, recruiting process, etc

Social planner picks # vacancies to maximize output $\Rightarrow$ picking tightness to maximize output.

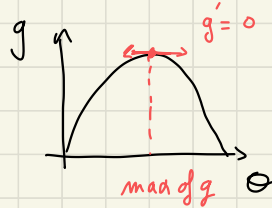
Efficient unemployment rate is unemployment rate chosen by social planner

Solution to social planner's problem

$$\max_V Y = a \underline{N^\alpha}$$

$$\begin{aligned} \Rightarrow \max_{\theta} N(\theta) &\rightarrow \begin{cases} \text{picking } V \Rightarrow \text{picking } \theta \\ \text{workers} = \text{producers} + \text{recruiters} \\ L = [1 + \tau(\theta)] N \\ \quad \quad \quad \uparrow \\ \quad \quad \text{recruiter/producer} \end{cases} \\ \Rightarrow \max_{\theta} \frac{L^S(\theta)}{1 + \tau(\theta)} &\} g(\theta) \end{aligned}$$

Necessary condition for maximum $\frac{dg}{d\theta} = 0$



$$\Rightarrow \frac{\theta}{g} \cdot \frac{dg}{d\theta} = 0 \quad \Rightarrow \quad \frac{d \ln g}{d \ln \theta} = 0 \quad \Rightarrow \quad \boxed{\varepsilon_g^\theta = 0}$$

$$\sum_{\theta} g = \sum_{\theta}^{L^S} - \sum_{\theta}^{1+\tau} \quad \leftarrow \eta \cdot \tau(\theta)$$

$$(1-\eta) u(\theta)$$

$$L^S = \frac{f(\theta)}{s+f(\theta)} H$$

$$\sum_{\theta}^{L^S} = \sum_{\theta} \frac{f(\theta)}{s+f(\theta)}$$

$$\left[\sum_{\theta} g = (1-\eta) u(\theta) - \eta \tau(\theta) \right]$$

Solution to social planner's problem θ that maximizes social welfare = output = # of producers is given by

$$(1-\eta) u(\theta) = \eta \tau(\theta)$$

$$\frac{u(\theta)}{\tau(\theta)} = \frac{\eta}{1-\eta}$$

$\rightarrow$ efficient labor market tightness
 θ^*

$\rightarrow$ efficient unemployment rate $u^* = u(\theta^*) = \frac{s}{s+f(\theta^*)}$

$\rightarrow$ efficient level of output = $a N(\theta^*)$
 $= a \left[\frac{L^S(\theta^*)}{1+\tau(\theta^*)} \right]^{\lambda}$

Application to the US labor market

Research on matching function (Petrongola & Pissanes, 2001).

$$\eta = 0.5 \Rightarrow \frac{\eta}{1-\eta} = 1$$

In practice

labor market is efficient when

$$\frac{u}{\tau} = 1$$

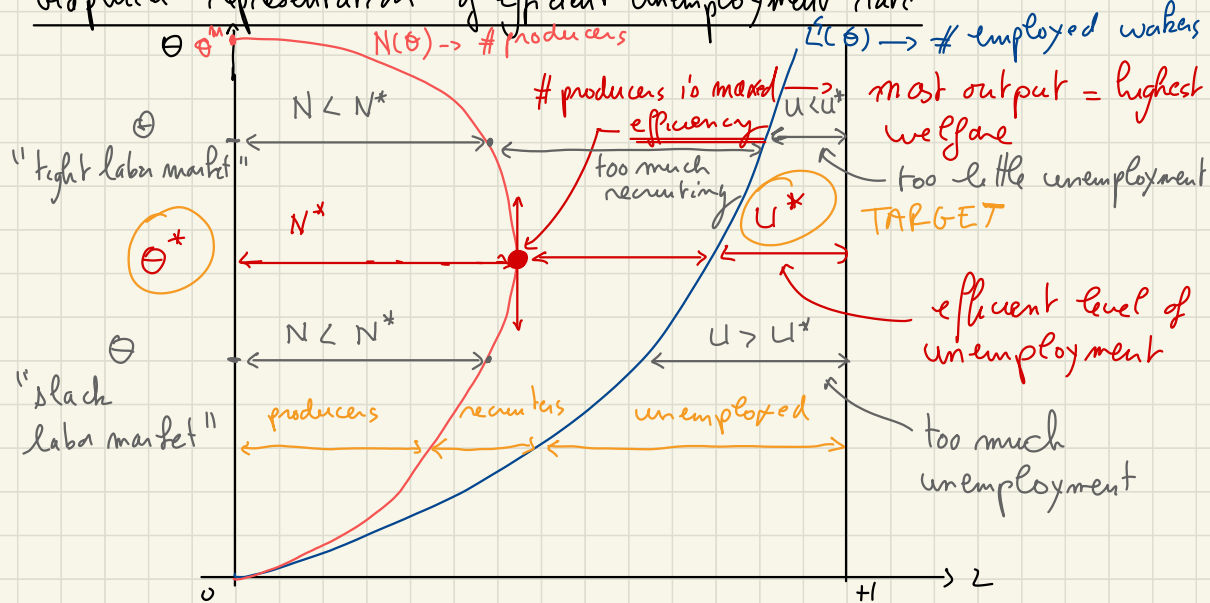
Rule of thumb.

unemployed workers = # recruiters

idle ↙

↗ non-productive

Graphical representation of efficient unemployment rate



In real world wage function may not guarantee that

$\theta = \theta^* \rightarrow$ firms may not have the incentive to post a # of vacancies such that

$\theta = \theta^*$ and $U = U^*$ and so on $\rightarrow$

government intervention may be needed to bring labor market closer to efficiency.

Efficient unemployment rate in Beveridgean models

Assumption Model admits a Beveridge curve

$$v = v(u)$$

where the function $v(u)$ is decreasing, convex

Examples

- Matching model
- Mismatch model
- Stock-flow matching model

Advantage Many countries have a Beveridge curve $\rightarrow$ method is applicable

Two key parameters.

- recruiting cost τ workers/vacancy
- social value of unemployment time / employment time z
 $z > 0$ or $z < 0$.

Social welfare $SW = (H - U) - \tau \times v + z L$

$\underbrace{(\# \text{ employed workers} - \# \text{ recruiters})}_{\text{producers}}$ $\underbrace{+ z L}_{\text{"output from unemployment"}}$

= producers

= output

"output from unemployment"

$$\text{Social welfare / capita} = \frac{SW}{H} = (1-u) - v \times r + z \cdot u$$

$$L \quad sw(u) = (1-u) + z \cdot u - v(u) \times r$$

Efficient unemployment rate u^* maximizes

$$sw(u) = \underbrace{1 - (1-z)u}_{\text{linear}} - \underbrace{v(u) \times r}_{\text{convex}} \quad | \quad \text{concave function}$$

Necessary condition for a maximum of the social welfare function is $\frac{dsw}{du} = 0$ (first-order condition)

Since $sw(u)$ is concave necessary condition is also sufficient $\rightarrow$ any u such that $dsw/du = 0$ is a maximum (maximum will be unique)

Efficient unemployment rate satisfies

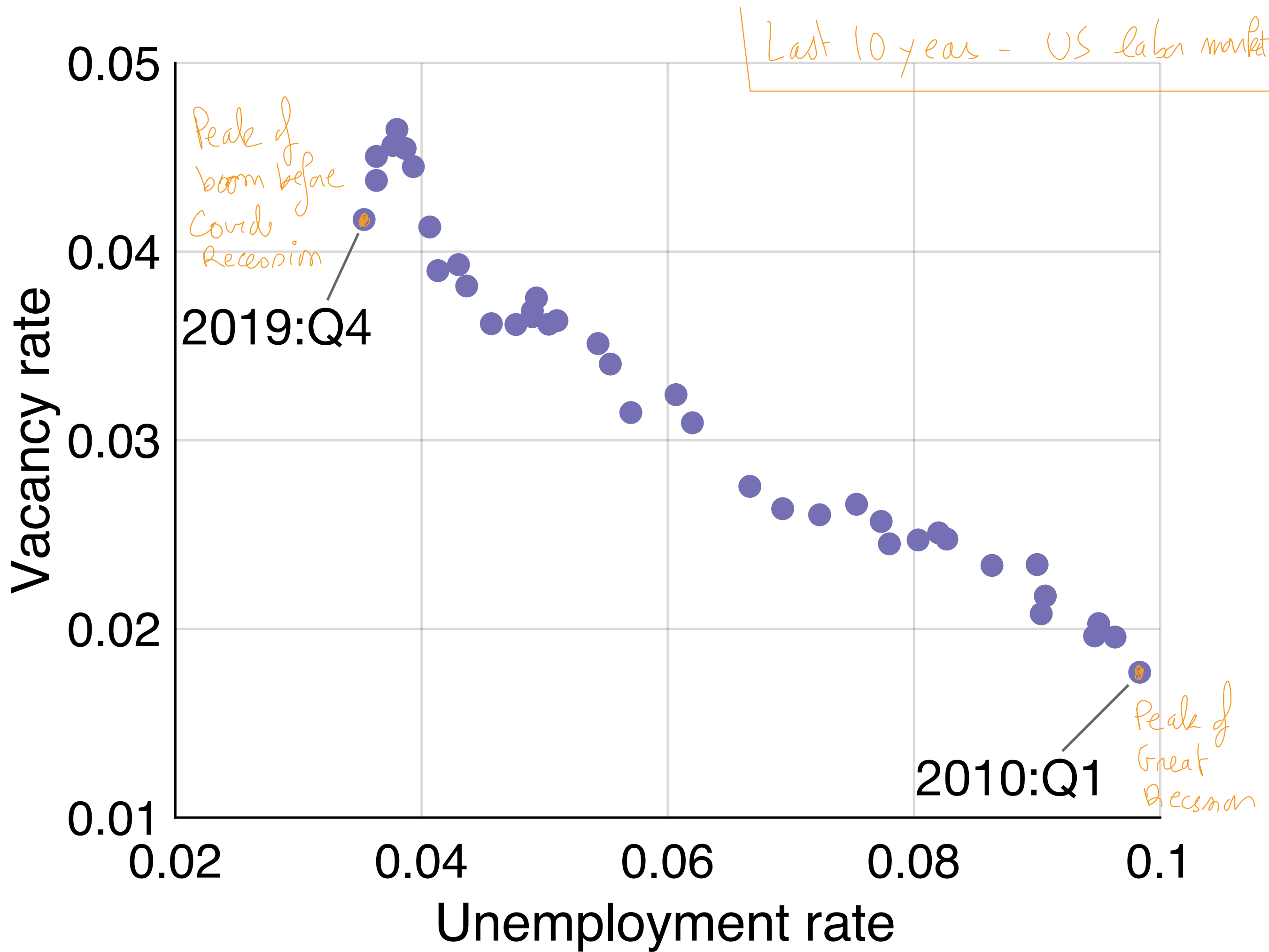
$$\frac{dsw}{du} = - (1-z) - v'(u) \times r = 0$$

$$\Rightarrow v'(u^*) = - \frac{1-z}{r}$$

• Slope of Beveridge curve = $- (1-z)/r$

• $v(u)$ is convex $\rightarrow v'(u)$ is strictly increasing $\rightarrow$ efficient unemployment rate is unique

• $r \uparrow \Rightarrow u^* \uparrow$ • $z \uparrow \Rightarrow u^* \uparrow$



Estimated

Beveridge curve

Efficiency

Slope of Beveridge curve = $-\frac{(1-z)}{r}$

Tangency point = Efficiency

isowelfare line

$$v = -\frac{1-z}{r} u + k$$

$$\Rightarrow -(1-z)u - v \cdot r = -kr$$

$$\Rightarrow 1 - (1-z)u - vr = k$$

$$\Rightarrow \boxed{sw(u) = k}$$

Higher welfare

$$-\frac{(1-z)}{r}$$

$$-\frac{(1-z)}{r}$$

0.05
0.04
0.03
0.02
0.01

0.02

0.04

0.06

0.08

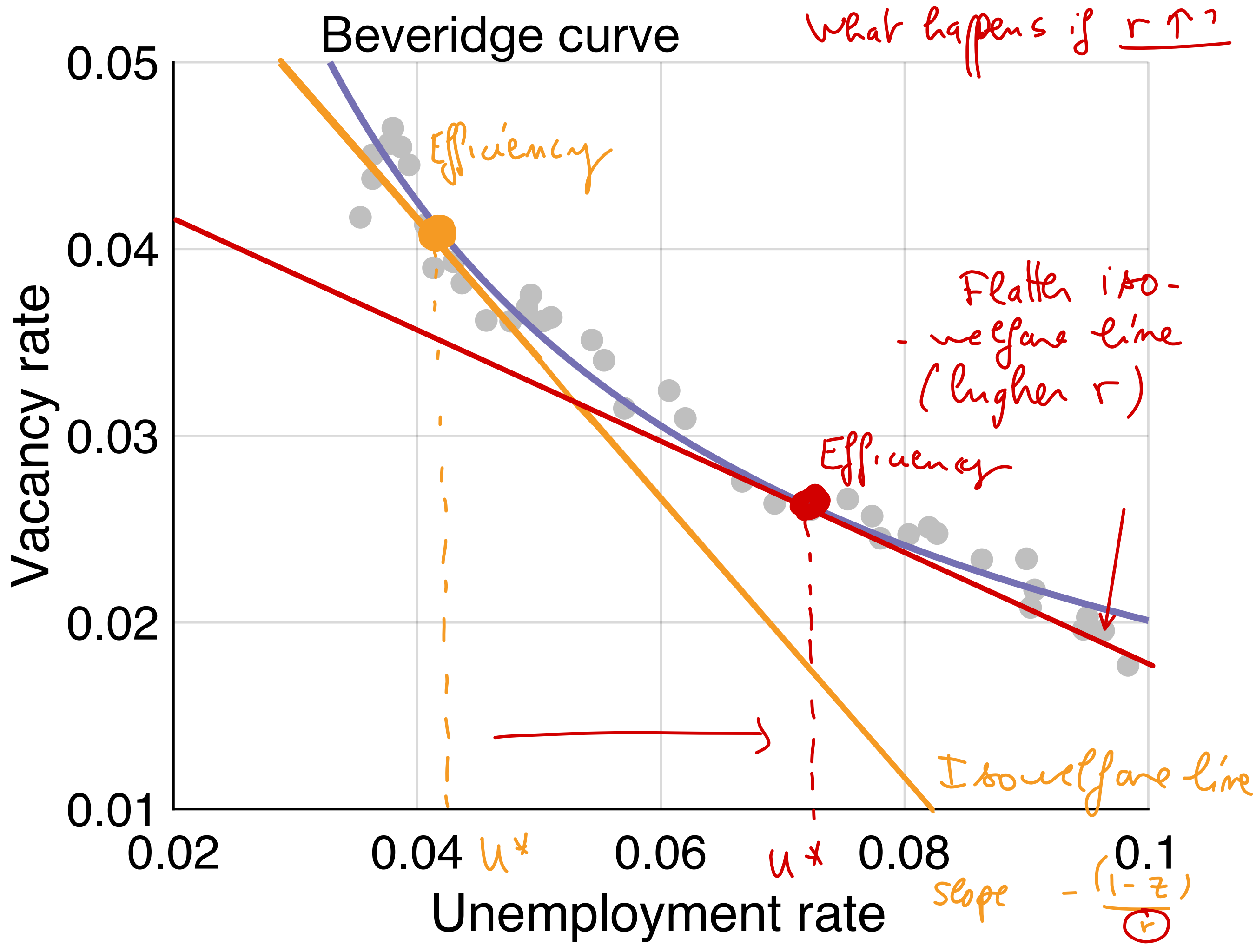
0.1

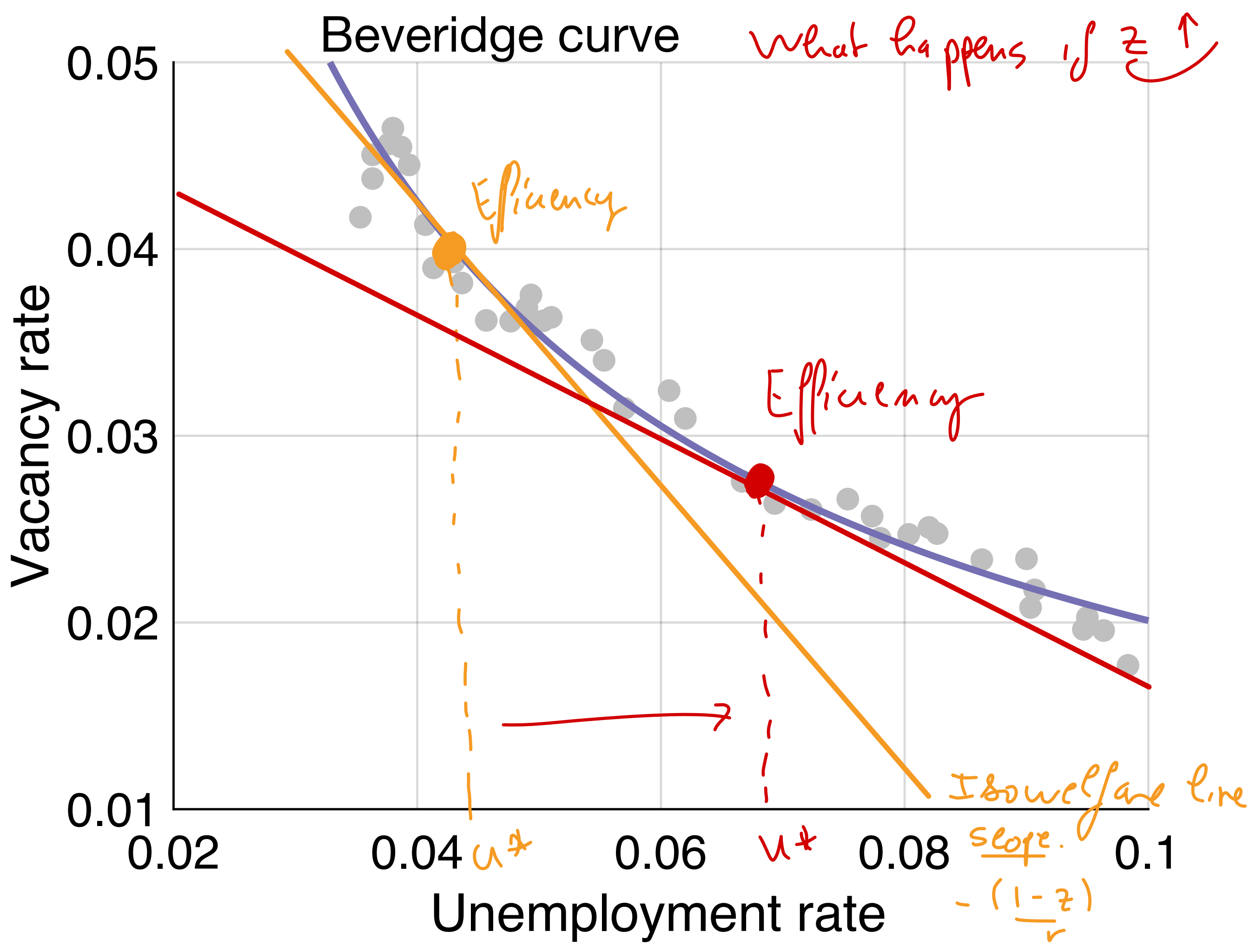
Unemployment rate

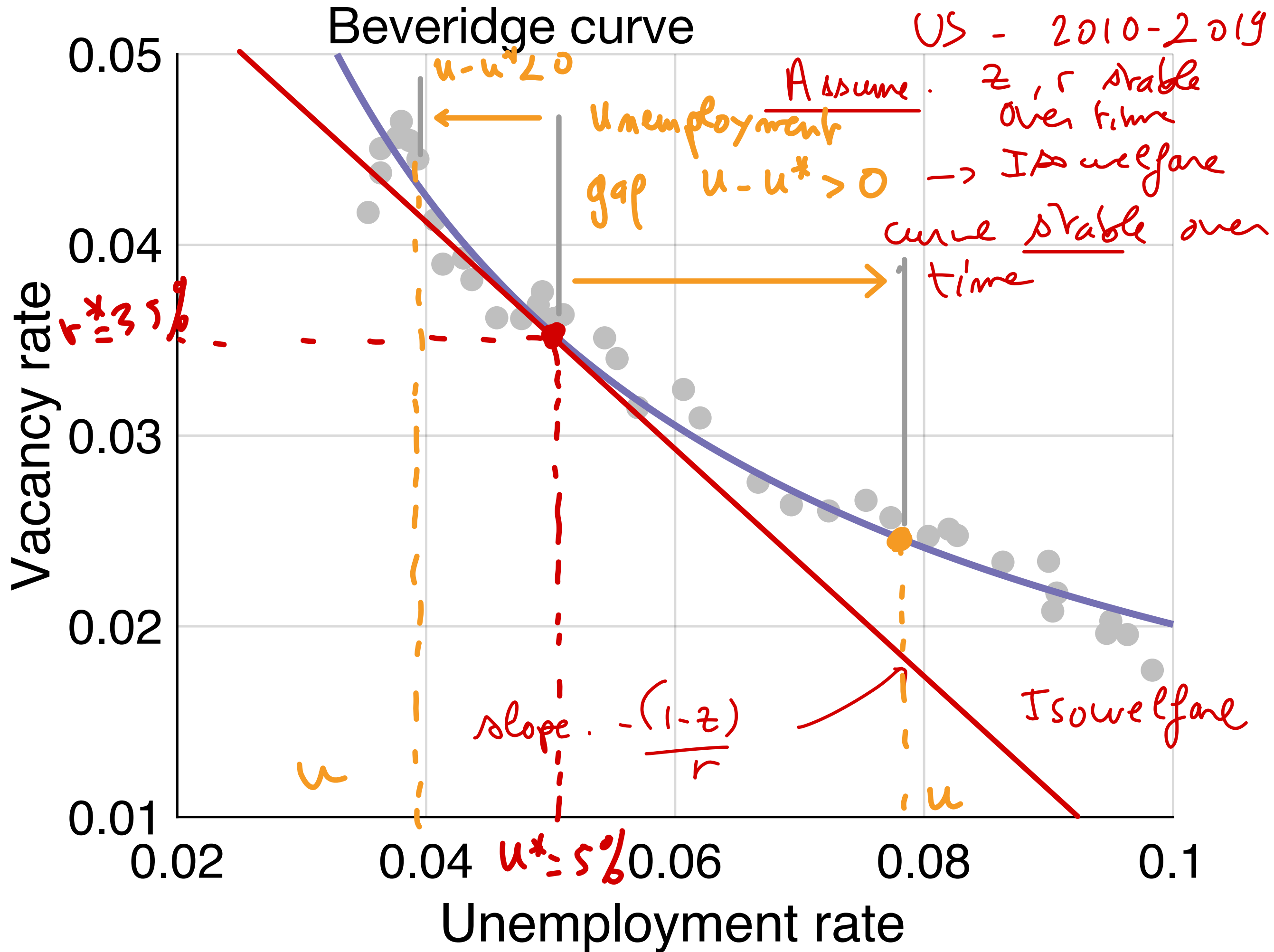
Vacancy rate

v^*

u^*







Formula for the efficient labor market tightness

Condition for labor market efficiency:

$$v'(u^*) = - \frac{1-z}{r}$$

Tightness $\Theta = v/u$

Beveridge elasticity
(normalized to be positive)

$$\varepsilon = - \frac{d \ln v}{d \ln u}$$

$$\varepsilon = - \frac{u}{v} \cdot \frac{dv}{du} = - \frac{v'(u)}{\Theta}$$

Efficiency condition

$$-\Theta v'(u) = \frac{1-z}{r}$$

Efficiency condition

$$\Theta^* = \frac{1-z}{\varepsilon r}$$

3 key factors

- z value of unemployment

$$z \uparrow \Rightarrow \Theta^* \downarrow, u^* \uparrow$$

- r recruiting cost

$$r \uparrow \Rightarrow \Theta^* \downarrow, u^* \uparrow$$

- ε elasticity of Beveridge curve

$$\varepsilon \uparrow \Rightarrow \Theta^* \downarrow, u^* \uparrow$$

Application to US labor market

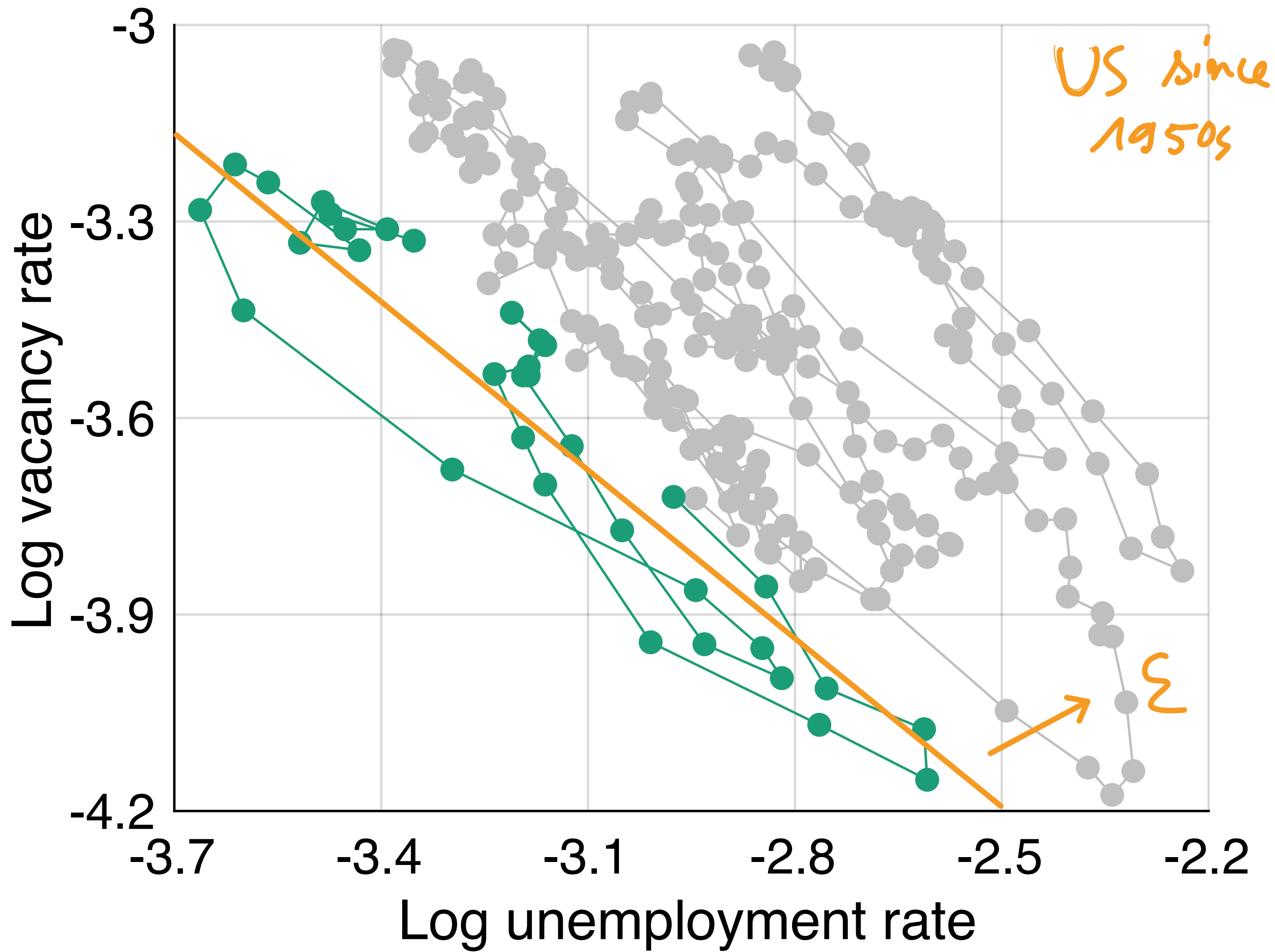
- 25% of labor costs devoted to recruiting (US 1997)
↳ 25% of workers are recruiters

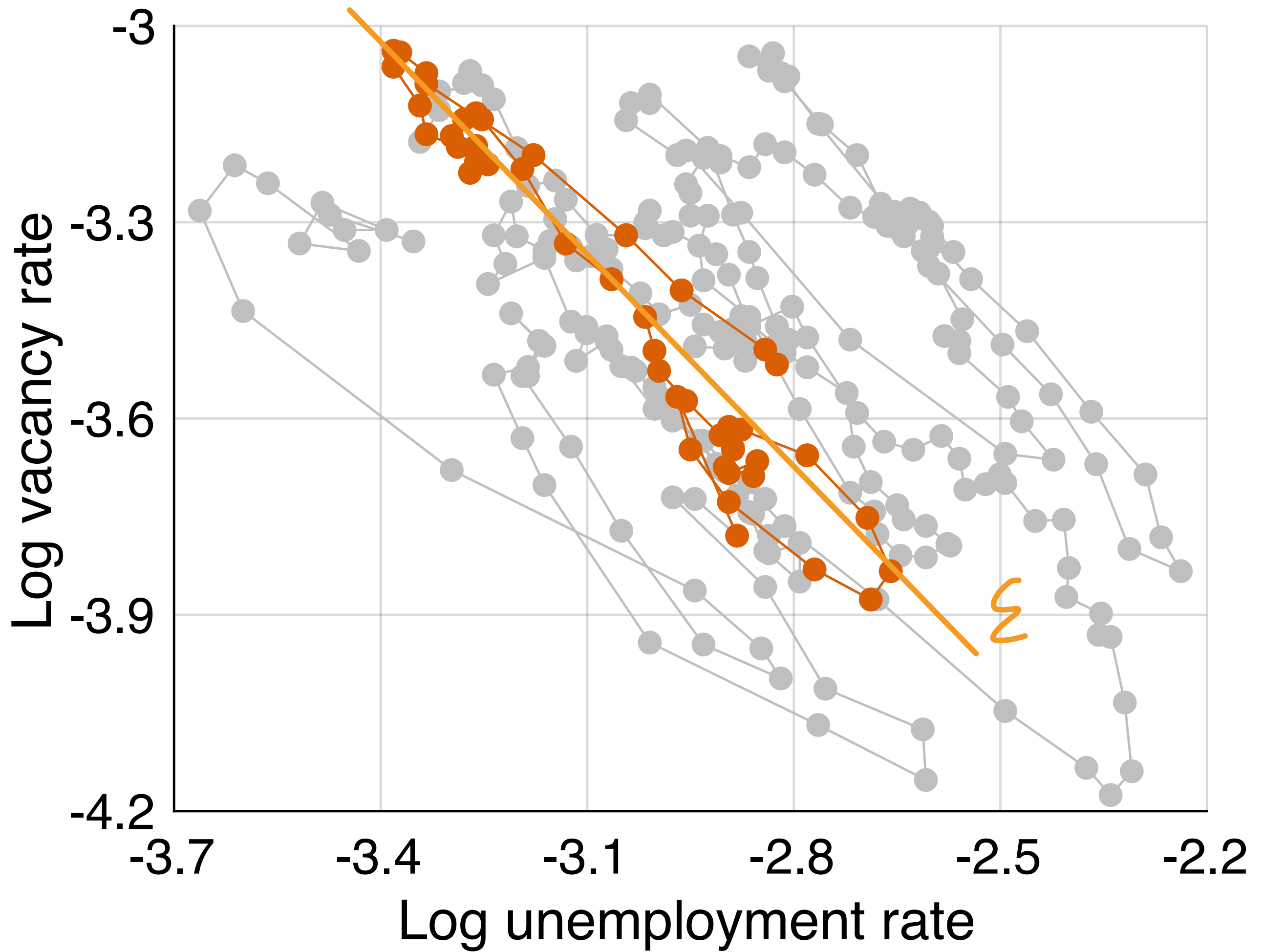
$$\tau \approx 0.7$$

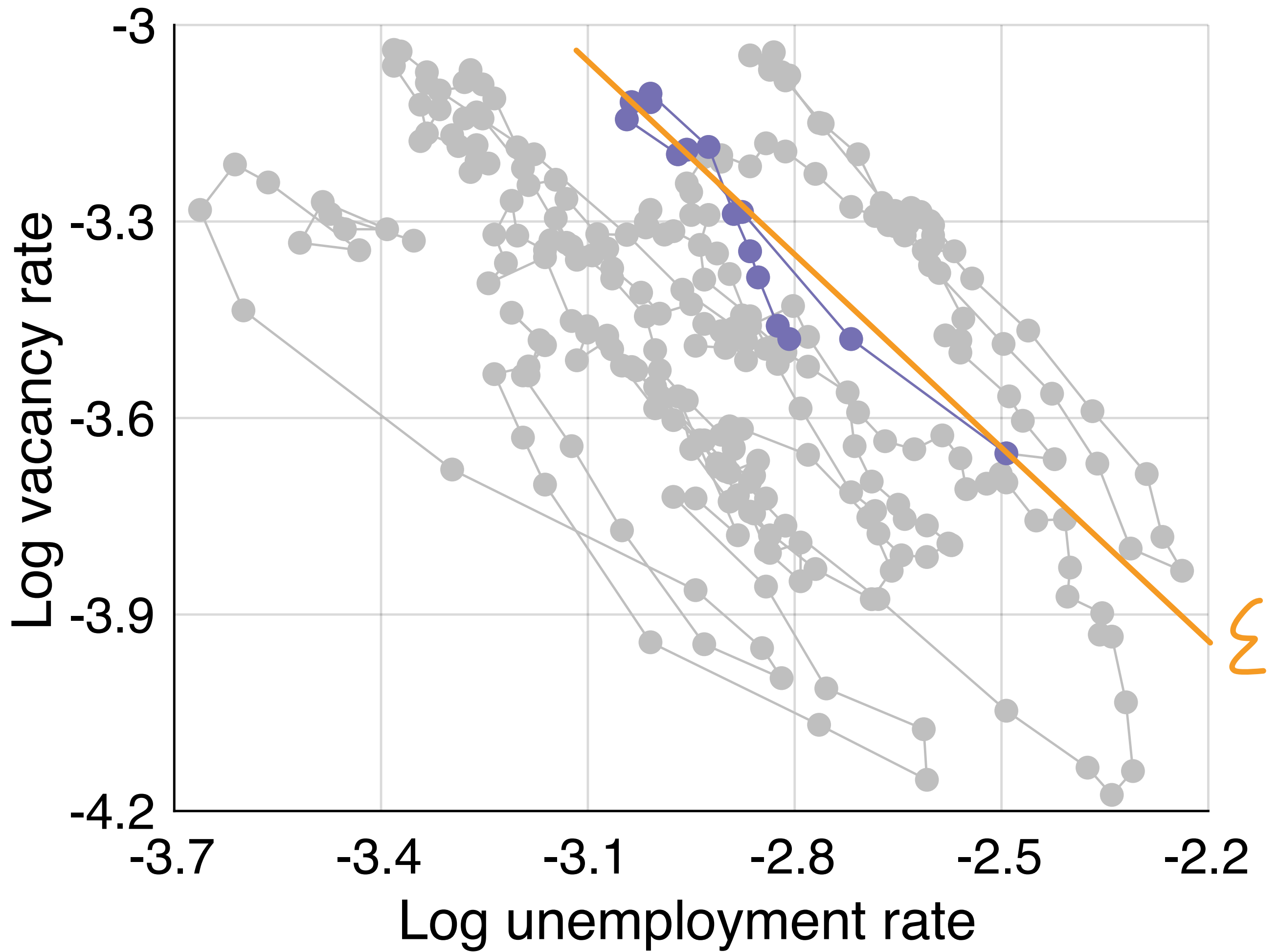
- 13% - 35% of total earnings ($\approx$ labor productivity) replaced by leisure & home production
 $13\% \leq \tau \leq 35\% \rightarrow \tau \approx 1/4$

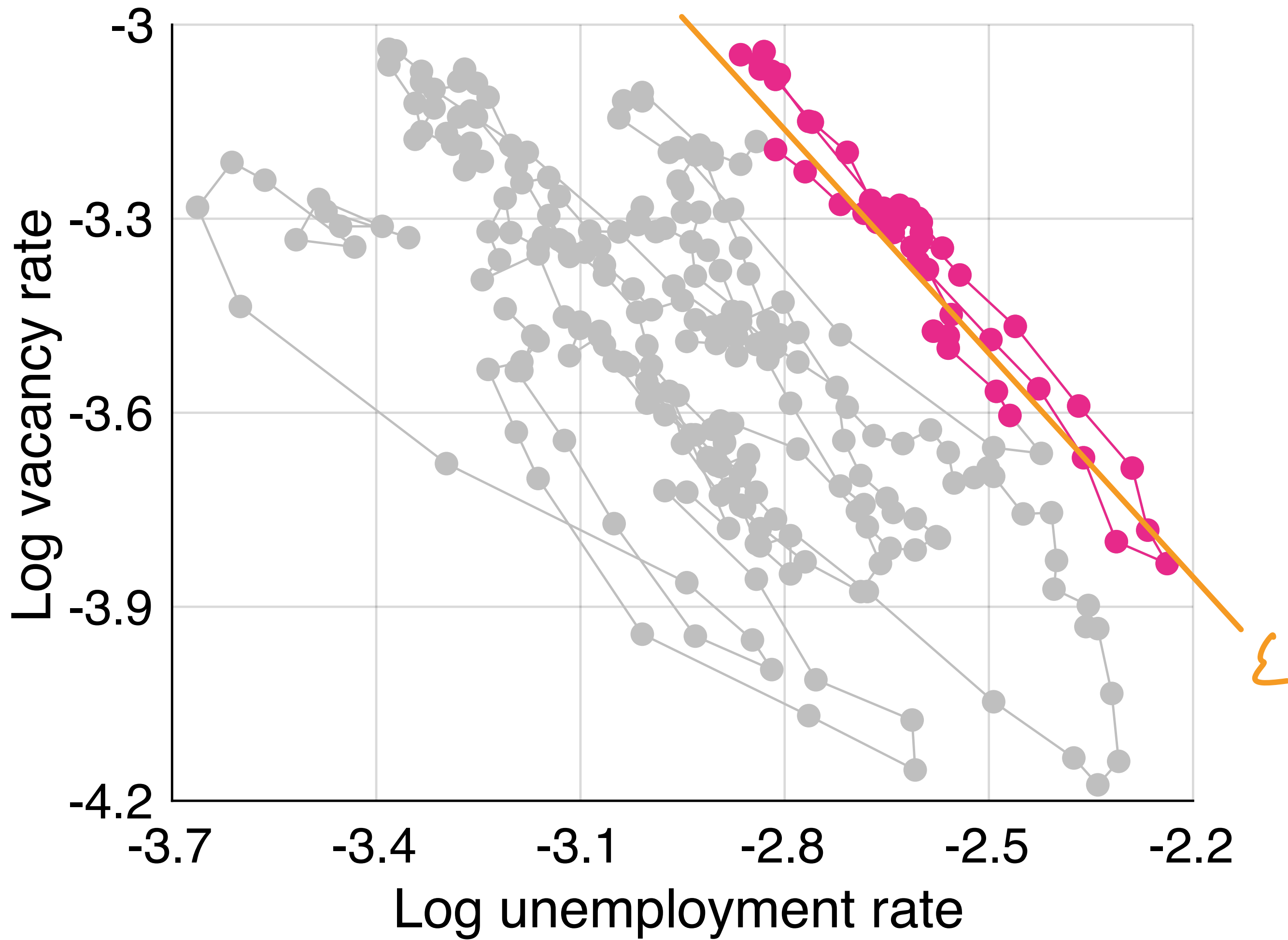
$$\varepsilon = \frac{d \ln v}{d \ln u}$$

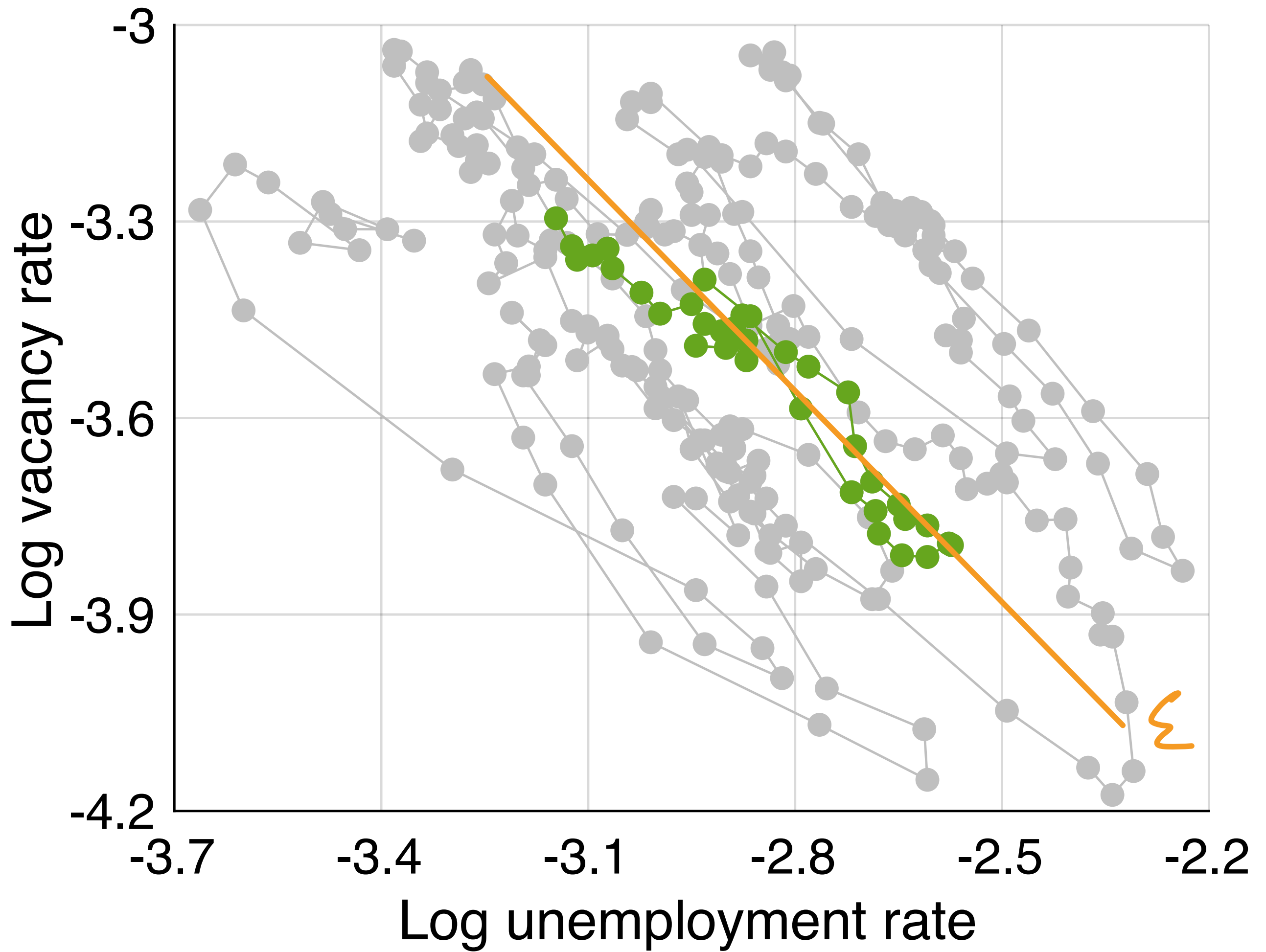
Slope of curve: $\ln v$ versus $\ln u$.
Coefficient in regression.

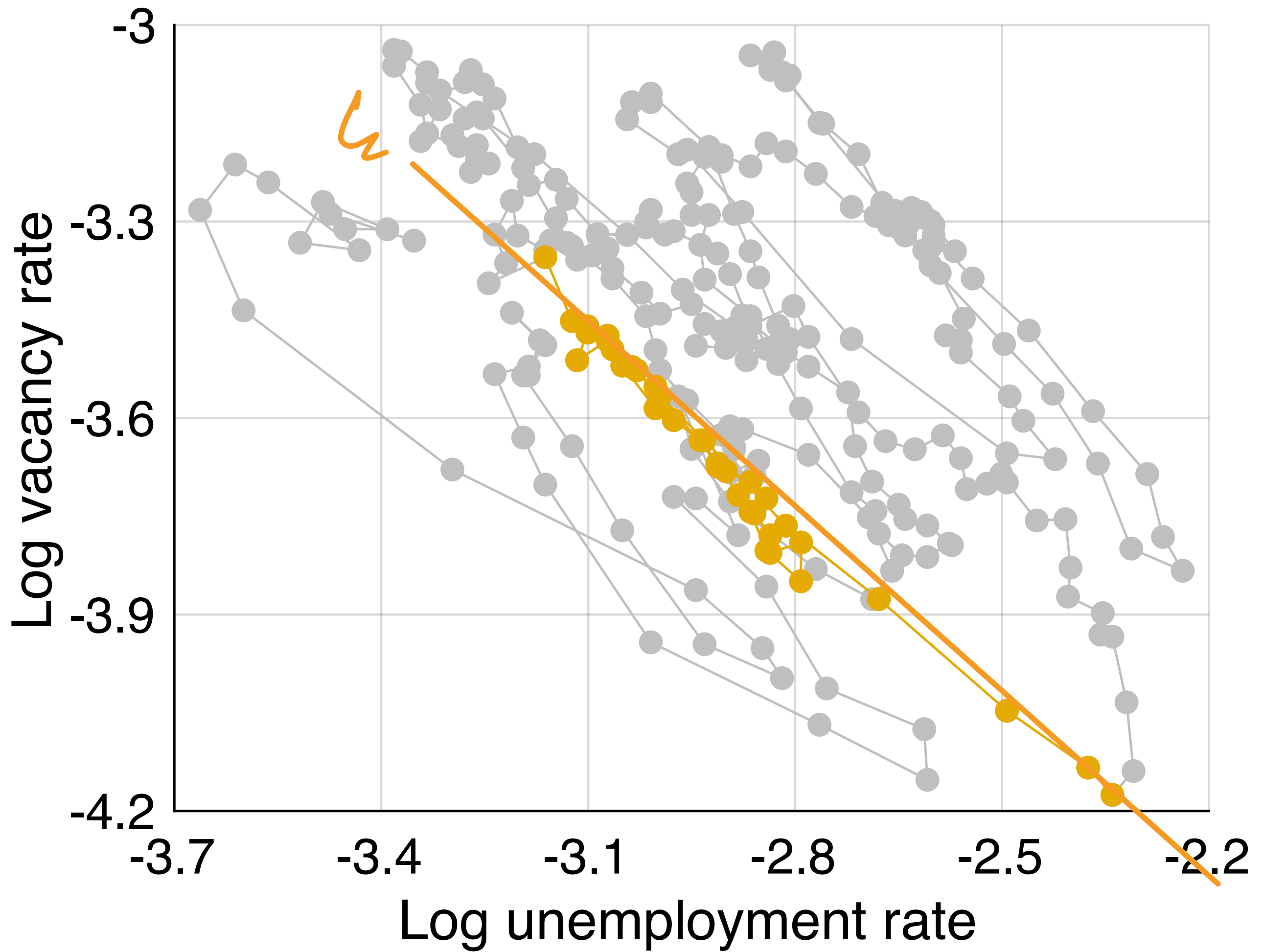


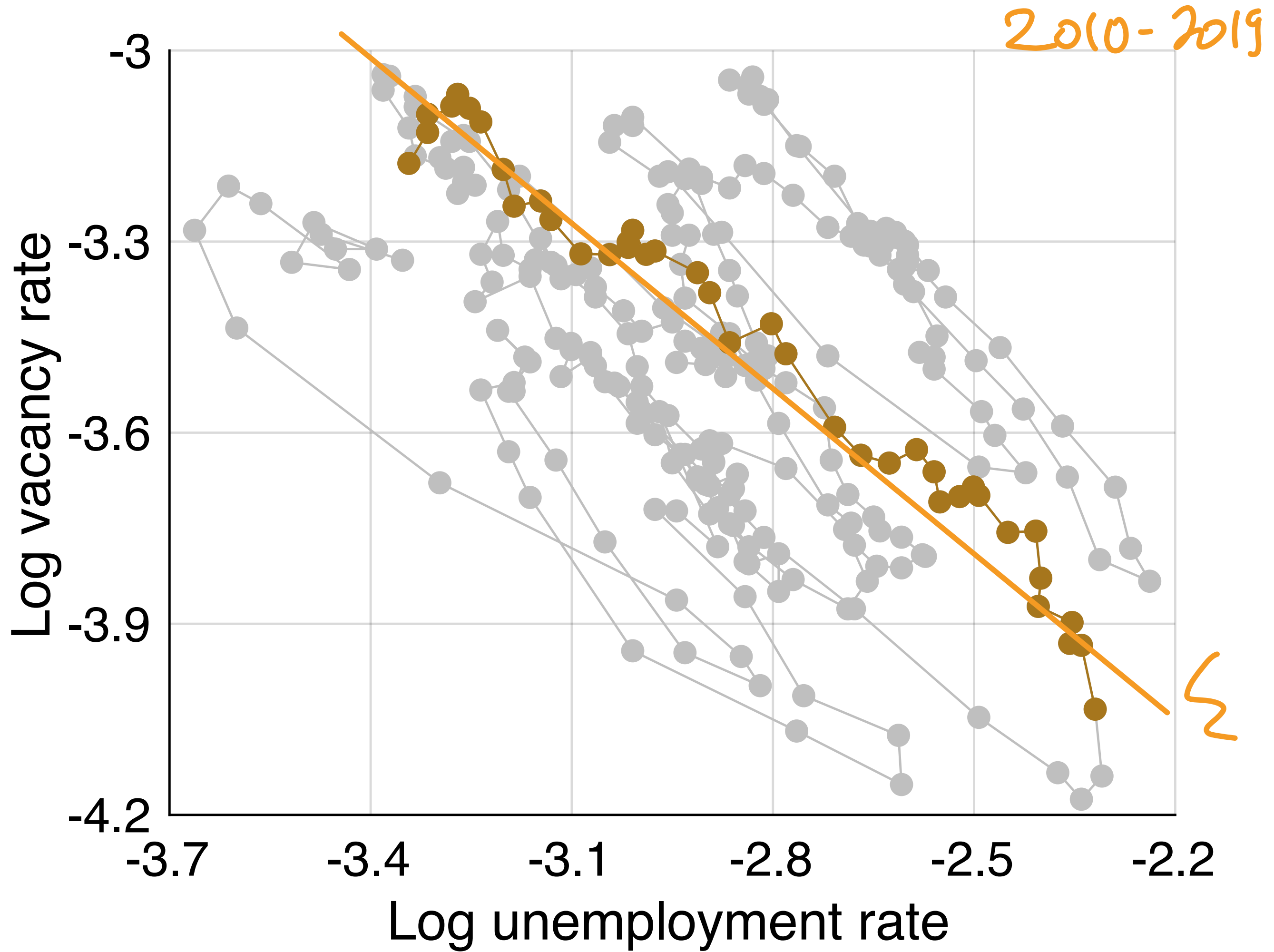


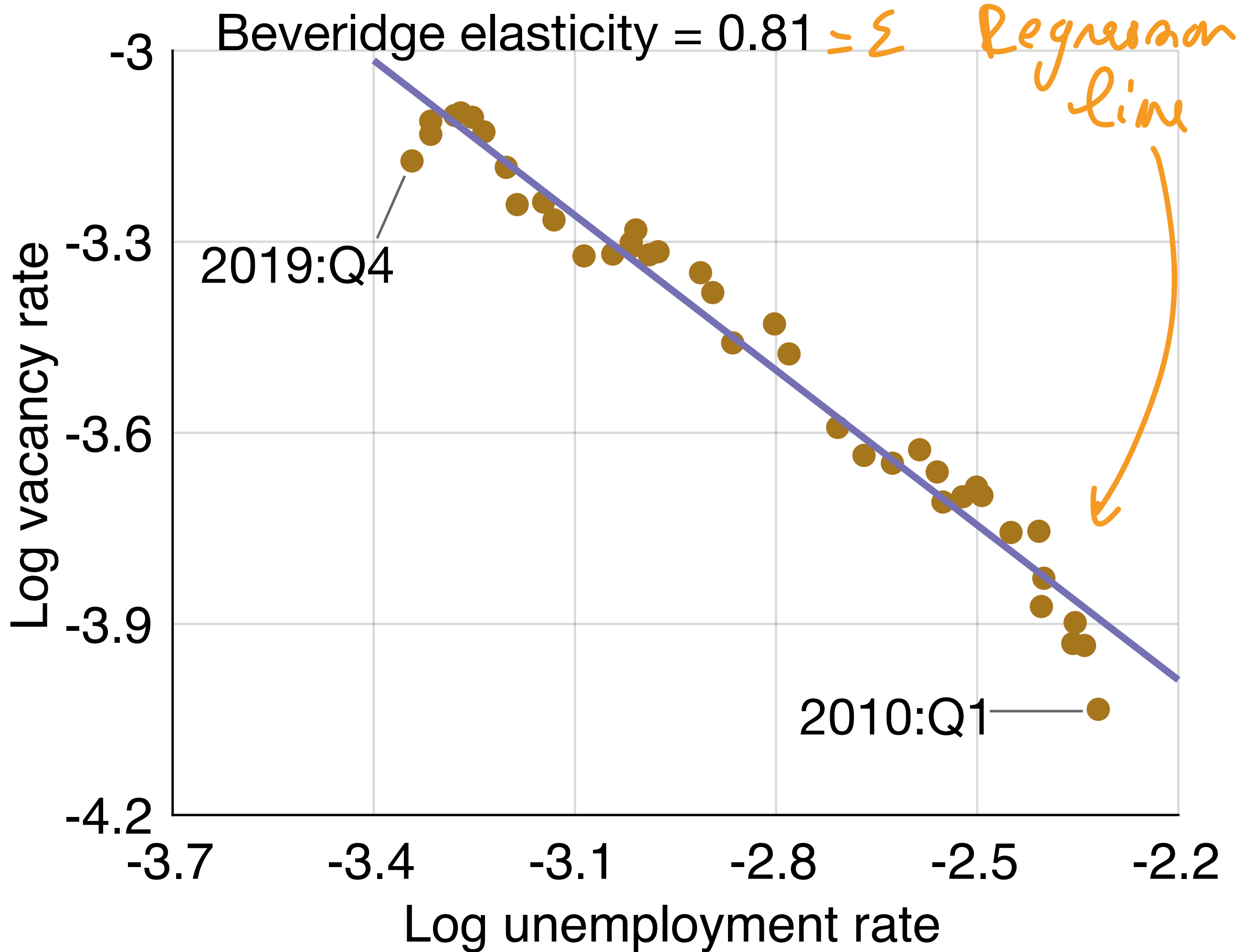


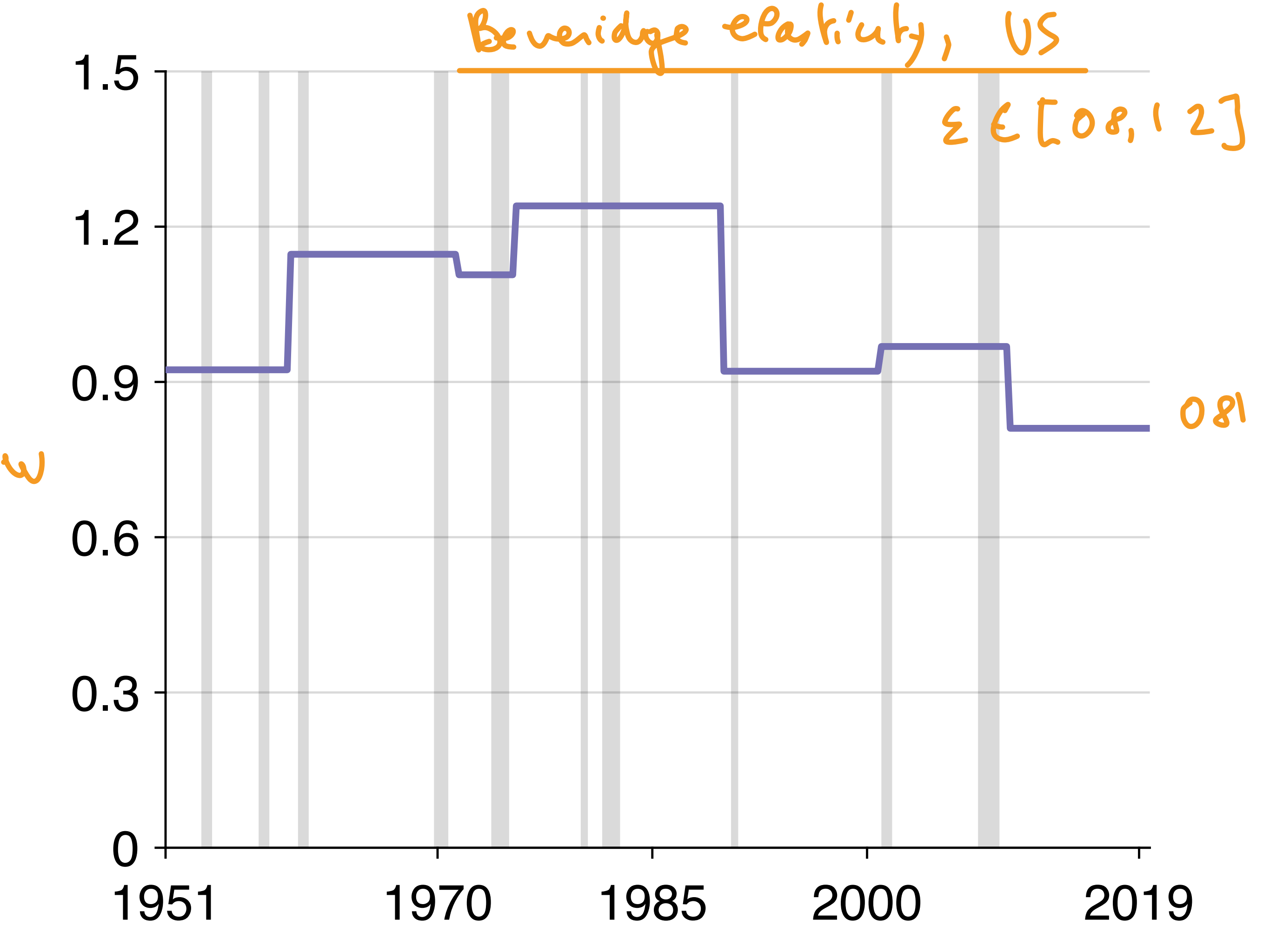












Unemployment rate

12%
9%
6%
3%
0%

1951

1970

1985

2000

2019

$\varepsilon, \tau, z \rightarrow \theta^* \rightarrow u^*$

US

Great Recession

Actual
 u

unemployment
GAP > 0

$u \approx u^*$

Vietnam
war

$u < u^*$

Korea war

Efficient
 u^*

